Course name: Data Communication LAB [K], Course code: COE 3201

Semester: Spring 2025-2026, Name of the Faculty: Dr. Muhammad Morshed Alam

Experiment no: 03

Experiment name: Study of Nyquist bit rate and Shannon capacity using MATLAB

Group no: 05

SL	Student name	ID	Marks
1	Maisha Mahjabin	23-54978-3	
2	Israt Jahan Surovy	23-54972-3	
3	Chowdhury Zarin Tasnim Oishy	23-54720-3	
4	Tanjimul Hossain Mojumder	23-52826-2	
5	Showmikul Islam Lemo	23-52830-2	

Marking Rubrics (to be filled by Faculty):

Category	Proficient [5]	Good [4]	Acceptable [3-2]	Unacceptable [1]	Secured Marks
Objective/ related theory	• Explains all necessary and relevant theoretical background information, measures, and variables.	• Explains the important parts of theoretical background information, may be partially irrelevant or need more details.	• Explains some theoretical background information, but some information may be missing, irrelevant or inaccurate.	• A lot of information is missing, out of context and/or inaccurate.	
Procedure/code	• Working procedure or code is clearly presented and is supported by proper comments. • Methods are clearly written, including all steps in sufficient detail for the experiment to be repeated.	• Working procedure or code is given for the experiment to be sufficient. • Methods are correct but the steps may be lacking in detail, making the experiment hard to be repeated.	• Working procedure or code is missing some steps and/or contains some mistakes. • Simulation output has many problems.	• Working procedure or code is absent or missing major steps and/or contains mistakes. • Simulation output completely wrong.	

Experiment results/table/simulation output figures	• Clear, accurate diagrams and graphs are labeled neatly and accurately with excellent detail. • Simulated result meets all criteria; outcomes are described clearly and accurately.	• Diagrams and plots are included and are correctly labeled in brief, but there may be some lack of clarity. • Most criteria are met, but there may be some lack of clarity and/or incorrect information.	• Diagrams and plots are included and are labeled, minor mistakes may be present. • Results do not match exactly with the theoretical values and/or analysis is unclear.	• Needed plots/ diagrams are missing or are missing important labels. • Experimental results are missing or completely incorrect.	
Data Interpretation/ analysis/question answer/conclusion	• Interpretation and analysis of related outcomes (consequences and implications) are logical. • Any report questions are properly answered with detailed justification or calculations.	• Analysis is logically tied to information; some related outcomes are not clear. • Report questions are answered correctly but may be lacking detail or contain minor logical error.	• Analysis is inconsistently tied to some of the information discussed; related outcomes are oversimplified. • Report questions are answered but contain wrong information or major logical error.	• Only the data was reported, there is no analysis. • Report questions are missing or are completely wrong.	
Comments:				Total Marks (Out of 20):	

Objectives:

1. To use MATLAB for studying fundamental concepts of communication engineering.
2. To analyze and understand the Nyquist bit rate and its limitations in a communication system.
3. To study Shannon capacity and determine channel capacity limits in the presence of noise.
4. To solve communication engineering problems using MATLAB simulations.
5. To bridge theoretical knowledge of Nyquist and Shannon theorems with practical implementation through MATLAB.

Theory:

Nyquist Bit Rate: The Nyquist bit rate formula defines the theoretical maximum bitrate for a noiseless channel.

$$BitRate = 2 \times bandwidth \times log_2 L$$

In this formula, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the bit rate in bits per second.

Shannon capacity: Shannon capacity formula was introduced to determine the theoretical highest data rate for a noisy channel:

$$Capacity = bandwidth \times log_2(1 + SNR)$$

In this formula, bandwidth is the bandwidth of the channel, SNR is the signal-to-noise ratio, and capacity is

the capacity of the channel in bits per second.

Signal-to-noise ratio (SNR): To find the theoretical bit rate limit, we need to know the ratio of the signal power to the noise power. The signal-to-noise ratio is defined as

$$\text{SNR} = \frac{Average\ Signal\ Power}{Average\ Noise\ Power}$$

We need to consider the average signal power and the average noise power because these may change with time.

A high SNR means the signal is less corrupted by noise; a low SNR means the signal is more corrupted by noise.

Because SNR is the ratio of two powers, it is often described in decibel units, SNR_{dB}, defined as

$$SNR_{dB} = 10 log_{10}(SNR)$$

Code:

a)

```
clc;
clear;

%%ID (23-54978-3)
A = 2;
B = 3;
C = 5;
D = 4;
E = 9;
H = 3;

%%(a) Select the value of the amplitudes
A1 = A + B + H;    % 8
A2 = B + C + H;    % 11
s  = (C + D + H)/30; % 12/30
```

b)

```
%%(b)Calculate the SNR value of the composite signal
fs = 8000;
t  = 0:1/fs:1-1/fs;

f1 = (C + D + H)*100;        % 1200 Hz
f2 = (D + E + H)*100;        % 1600 Hz

x = A1*sin(2*pi*f1*t) + A2*cos(2*pi*f2*t)+s*randn(size(t));

signal_power = (A1^2)/2 + (A2^2)/2;
noise_power  = s^2;
SNR          = signal_power / noise_power;
SNR_dB       = 10*log10(signal_power / noise_power);

fprintf('(b) SNR = %.4f dB\n', SNR_dB);
```

c)

```
%%(c)Find the bandwidth of the signal and calculate the maximum capacity of the channel
BW = abs(f1 - f2);
snrlin = 10^(SNR_dB/10);
Capacity = BW * log2(1 + snrlin);

fprintf('(c) Bandwidth = %.2f Hz\n', BW);
fprintf('    Maximum Channel Capacity  = %.2f bps\n', Capacity);
```

d)

```
%%(d) What will be the signal level to achieve the data rate?
R = 1000;
% Optimal signal levels from Nyquist formula
L = 2^(Capacity/(2*BW));
% Nyquist bit rate
BR = 2 * BW * log2(L);
fprintf('(d) Optimal Signal Levels (L) = %.4f\n', L);
fprintf('    Nyquist Bit Rate = %.2f bps\n', BR);

% Consistency check
if (Capacity <= BR)
    disp('Correct: Nyquist bit rate supports the channel capacity');
else
    disp('Not correct: Increase signal levels or bandwidth');
end
```

Result analysis/discussion:

Result:

```
(b) SNR = 27.6202 dB
(c) Bandwidth = 400.00 Hz
    Maximum Channel Capacity  = 3671.09 bps
(d) Optimal Signal Levels (L) = 24.0650
    Nyquist Bit Rate = 3671.09 bps
Correct: Nyquist bit rate supports the channel capacity
```

Discussion:

The bit rate of a channel depends on bandwidth, SNR, and signal levels. Bandwidth can be compared to the width of a road; more lanes allow more data to flow. SNR shows how clear the signal is compared to noise; a higher SNR allows faster and more reliable transmission. Signal levels are the number of distinct symbols used as more levels mean more bits per symbol, but they need a cleaner channel (high SNR) to avoid errors. In short, bandwidth sets the speed limit, SNR ensures clarity, and signal levels decide how much information fits in each signal.

Conclusion:

In conclusion, this experiment provides a robust platform for understanding and applying key concepts in communication engineering. Using MATLAB, we can analyze and manipulate parameters to assess the impact on communication system performance. By completing this experiment, we explored how varying parameters affect bit rate and channel capacity, gained practical insights into communication system design and optimization.

References:

[1] MathWorks, MATLAB Documentation, Natick, MA, USA: MathWorks. Online. Available: *https://www.mathworks.com/help/*

[2] Stormy Attaway, MATLAB: A Practical Introduction to Programming and Problem Solving, 6th ed. Boston, MA, USA: Butterworth-Heinemann, 2019.